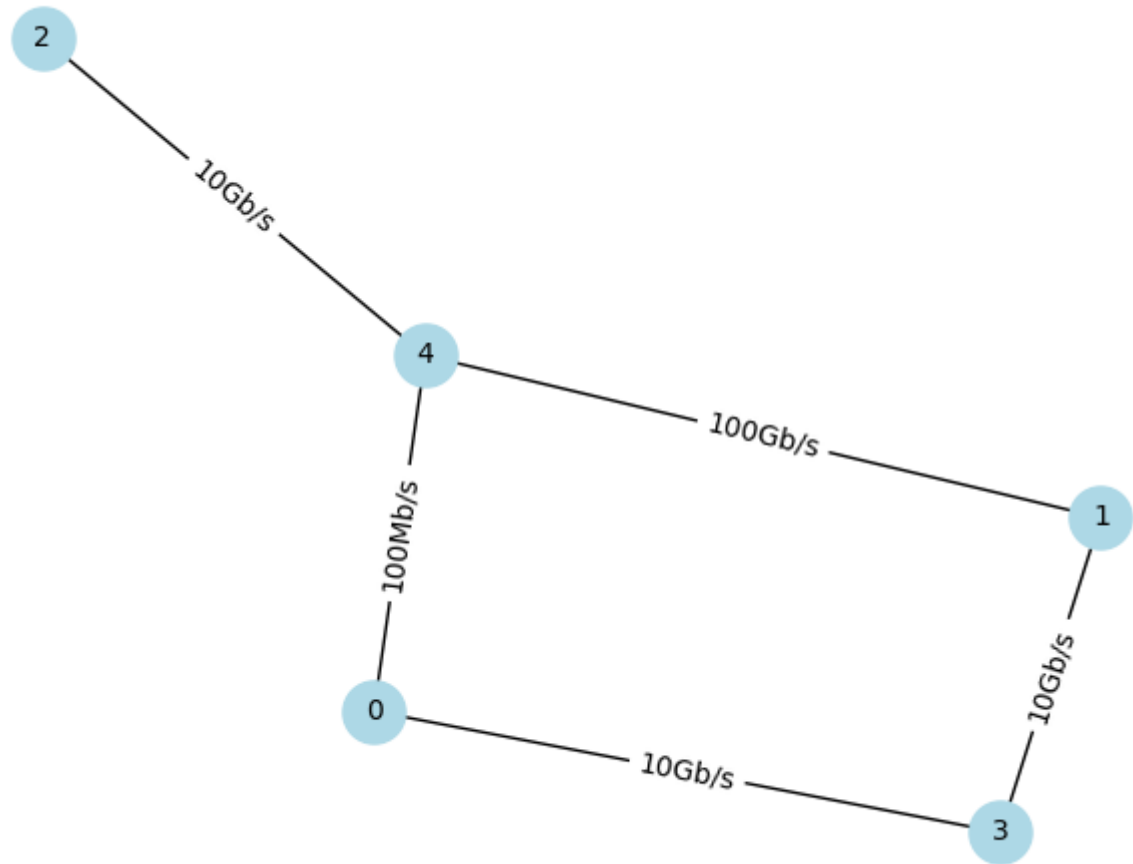


Spanning Tree Protocol

Seed: 189026065



Consider the network in the figure and assume that:

- The adopted algorithm is the **Spanning Tree Protocol (STP)**.
- BPDUs are formatted as a tuple: (Root ID, Path Cost, Sender ID).
- Tie-breaking preference is: **Lowest Root ID, Lowest Cost, Lowest Sender ID**.
- For simplicity we don't add the ports to the BPDUs, so assume they are generated in node order (from zero on) and they are received in neighbors order. The solution assumes a neighbor with label X is on port X.
- Every port is initialized in an undefined state, and every port must be assigned for the protocol to converge

Instructions:

Write down the sequence of message exchanges. For every BPDU that causes a state change in the receiver, write:

1. The **Sender -> Receiver** and the **[BPDU]** content.
2. The receiver's **Updated Best BPDU**.
3. The receiver's **Updated Port List** using the format **[Neighbor:ROLE]**.

Use the following abbreviations for roles: ROOT, DESI (Designated), BLOC (Blocked), and UNDE (Undefined).

Message Sequence

0 -> 3: [BPDU] (0, 0, 0)

Current BPDU of 3: (3, 0, 3)
Computed Priority Vector: (0, 2000, 0)
Updated BPDU of 3: (0, 2000, 3)
Updated Ports of 3: [0:ROOT, 1:UNDE]

0 -> 4: [BPDU] (0, 0, 0)

Current BPDU of 4: (4, 0, 4)
Computed Priority Vector: (0, 200000, 0)
Updated BPDU of 4: (0, 200000, 4)
Updated Ports of 4: [0:ROOT, 1:UNDE, 2:UNDE]

1 -> 3: [BPDU] (1, 0, 1)

Current BPDU of 3: (0, 2000, 3)
Computed Priority Vector: (1, 2000, 1)
Updated BPDU of 3: (0, 2000, 3)
Updated Ports of 3: [0:ROOT, 1:DESI]

1 -> 4: [BPDU] (1, 0, 1)

Current BPDU of 4: (0, 200000, 4)
Computed Priority Vector: (1, 200, 1)
Updated BPDU of 4: (0, 200000, 4)
Updated Ports of 4: [0:ROOT, 1:DESI, 2:UNDE]

2 -> 4: [BPDU] (2, 0, 2)

Current BPDU of 4: (0, 200000, 4)
Computed Priority Vector: (2, 2000, 2)
Updated BPDU of 4: (0, 200000, 4)
Updated Ports of 4: [0:ROOT, 1:DESI, 2:DESI]

3 -> 0: [BPDU] (0, 2000, 3)

Current BPDU of 0: (0, 0, 0)
Computed Priority Vector: (0, 4000, 3)
Updated BPDU of 0: (0, 0, 0)
Updated Ports of 0: [3:DESI, 4:UNDE]

3 -> 1: [BPDU] (0, 2000, 3)

Current BPDU of 1: (1, 0, 1)
Computed Priority Vector: (0, 4000, 3)
Updated BPDU of 1: (0, 4000, 1)
Updated Ports of 1: [3:ROOT, 4:UNDE]

4 -> 0: [BPDU] (0, 200000, 4)

Current BPDU of 0: (0, 0, 0)
Computed Priority Vector: (0, 400000, 4)
Updated BPDU of 0: (0, 0, 0)
Updated Ports of 0: [3:DESI, 4:DESI]

4 -> 1: [BPDU] (0, 200000, 4)

Current BPDU of 1: (0, 4000, 1)
Computed Priority Vector: (0, 200200, 4)
Updated BPDU of 1: (0, 4000, 1)
Updated Ports of 1: [3:ROOT, 4:DESI]

4 -> 2: [BPDU] (0, 200000, 4)

Current BPDU of 2: (2, 0, 2)
Computed Priority Vector: (0, 202000, 4)
Updated BPDU of 2: (0, 202000, 2)
Updated Ports of 2: [4:ROOT]

1 -> 4: [BPDU] (0, 4000, 1)

Current BPDU of 4: (0, 200000, 4)
Computed Priority Vector: (0, 4200, 1)
Updated BPDU of 4: (0, 4200, 4)
Updated Ports of 4: [0:BLOC, 1:ROOT, 2:DESI]

4 -> 2: [BPDU] (0, 4200, 4)

Current BPDU of 2: (0, 202000, 2)
Computed Priority Vector: (0, 6200, 4)
Updated BPDU of 2: (0, 6200, 2)
Updated Ports of 2: [4:ROOT]

Final Network State

Node 0

Cost: 0

Port State: [3:DESI, 4:DESI]

Node 1

Cost: 4000

Port State: [3:ROOT, 4:DESI]

Node 2

Cost: 6200

Port State: [4:ROOT]

Node 3

Cost: 2000

Port State: [0:ROOT, 1:DESI]

Node 4

Cost: 4200

Port State: [0:BLOC, 1:ROOT, 2:DESI]